

Atomistic Modeling of W/h-BN/Graphene Multilayer Coatings for the Protection of the First Wall of a Fusion Reactor: Concept Justification, Validation of Neural Network Potentials, and the First Stage of the Study

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Abstract

The paper considers the degradation of the first wall of a fusion reactor under the combined action of helium fluxes, hydrogen isotopes, and fast neutrons. The standard wall material, tungsten, is destroyed due to helium erosion, hydrogen embrittlement, and neutron cascades. A multilayer protective coating architecture is proposed: a tungsten substrate providing heat removal; a hexagonal boron nitride layer enriched with boron-10 intended for neutron absorption; and an outer graphene barrier limiting the penetration of helium and hydrogen isotopes.

The modeling was performed by molecular dynamics with neural network potentials in the GPUMD package. The universal NEP89 potential covering the elements W, B, N, and C was chosen as the main potential. Validation was carried out: the tungsten lattice parameter was reproduced with a deviation of 0%, and the C–C bond lengths in graphene and B–N bonds in h-BN were reproduced with deviations of 0.02% and 0.06%, respectively. It was established that the three-layer system remains structurally stable at 1200 K, and the layers do not mix. The adhesion energy of the W/h-BN interface is:

$$E_{\text{adh}} = -16.48 \text{ eV}, \quad (1)$$

which indicates stable interlayer bonding.

A separate test showed that NEP89 is unsuitable for describing the interaction of He with tungsten: the predicted insertion energy was:

$$E_{\text{ins}} = -41.7 \text{ eV}, \quad (2)$$

whereas the DFT value lies in the range of $+4 \dots +6$ eV. This result justifies the need to train a specialized potential for helium, which determines the next task of the study.

This work is the first stage of a single study, the results of which will be combined into a final article for a peer-reviewed journal.

Keywords

fusion reactor; first wall; tungsten; hexagonal boron nitride; graphene; neural network potentials; molecular dynamics; GPUMD; NEP89; helium erosion; potential validation; multilayer coatings; adhesion.

1 Introduction

Controlled nuclear fusion remains one of the most attractive goals of modern energy. Unlike nuclear fission, the fusion of light nuclei promises a practically inexhaustible energy source with significantly less long-lived radioactive waste. The international ITER project under construction in France is intended to demonstrate the technical feasibility of confining fusion plasma and obtaining a positive energy output. The subsequent DEMO project is already aimed at electricity production. However, between plasma physics and an industrial reactor stands a materials science wall that all projects stumble upon.

The first wall is the inner surface of the vacuum chamber directly facing the plasma. The magnetic field confines the plasma itself but cannot stop neutral particles and radiation leaving the confinement zone. A combined flux falls on the wall: helium ions produced in the deuterium–tritium reaction; hydrogen isotopes not retained by the magnetic field; fast neutrons with an energy of 14.1 MeV for which the wall is practically transparent; and intense thermal radiation, especially during plasma disruptions.

Tungsten has been chosen as the standard first-wall material. It has the highest melting point among metals (3422 °C), good thermal conductivity, and a high atomic displacement threshold of about 90 eV. It would seem to be an ideal candidate. But tungsten has an Achilles' heel: helium. Helium is almost insoluble in tungsten. Entering the lattice, its atoms do not distribute uniformly but seek free volumes: vacancies, pores, grain boundaries. There they accumulate, merge into bubbles, and create internal stresses. Blisters appear on the surface, and at high fluxes a nanostructured layer known as fuzz forms. At the same time, hydrogen isotopes penetrate defects and cause embrittlement, while neutrons generate displacement cascades and eventually transmute tungsten into rhenium and osmium.

Traditional approaches such as alloying, nanostructuring, and surface texturing only slow the degradation. They do not remove the cause: a single material is forced to resist three physically different threats at once.

In the present work, a different path is proposed: functional decomposition. Instead of one universal material, a multilayer coating is built in which each layer is responsible for its own threat. Tungsten remains the substrate and heat sink. Above it lies hexagonal boron nitride enriched with the boron-10 isotope, which efficiently absorbs neutrons. The outer layer is graphene, a two-dimensional carbon crystal that is practically impermeable to helium and hydrogen isotopes.

The aim of this work is to test the physical validity of this architecture by atomistic modeling. Several tasks must be solved: to select and justify a computational method; to generate and validate a model of the three-layer system; to check its thermal stability at the working temperature; to estimate interlayer adhesion; and to identify the limitations of the chosen approach.

2 Structure of the Study

This publication is not an isolated work. It opens a series of four consecutive studies united by a common goal: to determine whether the W/h-BN/graphene multilayer architecture can protect the first wall of a fusion reactor. Each part solves its own task, but all of them are stages of one project.

The first work presented here lays the foundation. It justifies the concept, selects the computational tool, validates the NEP89 neural network potential for the elements W, B, N, and C, checks the thermal stability of the three-layer system, and calculates the adhesion energy of the W/h-BN interface. The limitations of the approach, primarily those related to helium, are identified separately.

The second work will be devoted to creating a specialized neural network potential for helium and modeling helium erosion of the coating. The third will cover the modeling of the $^{10}\text{B}(n, \alpha)^7\text{Li}$ reaction and the combined action of helium, hydrogen isotopes, and neutrons. The fourth and final work will summarize the results, estimate the coating lifetime, and formulate practical recommendations.

After all stages are completed, the results will be combined into a single article for submission to a peer-reviewed journal. Only the first stage is published now, but it determines the entire further trajectory.

3 Literature Review

The first-wall problem is documented in detail in ITER materials and numerous reviews on plasma-material interactions. Helium erosion of tungsten has been studied most thoroughly. Experiments at the linear plasma devices PISCES and NAGDIS, as well as the works of S. Kajita, N. Ohno, and colleagues, have shown that irradiation of tungsten with helium ions produces blisters and, at high temperatures and fluxes, a fuzz nanostructure. The mechanism is well known: the low solubility of helium forces its atoms to cluster in vacancies and pores, and growing bubbles destroy the surface.

Hydrogen embrittlement has also been studied experimentally and theoretically. Deuterium is retained in lattice defects, and simultaneous irradiation with helium and hydrogen enhances bubble formation. The He-H synergy is recognized as one of the most dangerous degradation mechanisms in tungsten.

Neutron damage is associated with displacement cascades and the transmutation

of tungsten into rhenium and osmium. Studies performed for DEMO show that the accumulation of these elements leads to embrittlement and reduced thermal conductivity.

A separate direction is protective coatings based on two-dimensional materials. It has been experimentally established that graphene is impermeable to helium atoms and ions at energies up to several hundred eV. Hexagonal boron nitride has high thermal and radiation resistance. Boron-10, which is part of h-BN, has one of the largest thermal neutron capture cross sections among light elements, making this material a natural candidate for neutron shielding.

On the computational side, neural network potentials play a key role. The NEP method developed by Zheyong Fan and colleagues provides accuracy close to DFT while allowing the modeling of systems with thousands and tens of thousands of atoms. The universal NEP89 potential covers 89 elements, including tungsten, boron, nitrogen, and carbon.

The gap filled by the present work is as follows: no studies have been found in the available literature that combine tungsten, h-BN, and graphene into a single multilayer architecture and study it by molecular dynamics with neural network potentials. This combination of materials and computational method remains unexplored.

4 Hypothesis and Design Rationale

The central hypothesis of the work: a multilayer coating with separation of protective functions among layers will provide a longer lifetime of the first wall than uniform tungsten or separate thin-film protections.

Consider the nature of the three threats.

Helium is an inert gas. Its dissolution energy in tungsten is positive and large. Entering the lattice, a helium atom cannot distribute uniformly and is forced to seek voids. In voids, He atoms merge into bubbles, the pressure inside them grows, and the metal is destroyed from within. Hydrogen acts similarly, but additionally penetrates existing helium bubbles and enhances their growth. Neutrons have no charge and penetrate deeply into the material, creating displacement cascades and transmutation.

A single material cannot equally effectively resist all three threats. Therefore, different materials are needed.

Tungsten remains the substrate: it removes heat and carries mechanical load. Its refractoriness and thermal conductivity are irreplaceable. But it must not contact the plasma.

Graphene takes on helium and hydrogen. It is a two-dimensional crystal with high electron density, impermeable even to helium atoms. It is strong and thermally conductive in the plane.

h-BN with boron-10 absorbs neutrons. The reaction $^{10}\text{B}(n, \alpha)^7\text{Li}$ converts a neutron

into charged particles that are stopped locally and do not create deep cascades in tungsten.

There are also objections. First, h-BN conducts heat worse than tungsten and may worsen heat removal. Answer: the h-BN thickness is a few nanometers, and the thermal resistance is negligible. Second, during neutron capture, h-BN itself degrades because lithium and helium are born inside it. Answer: the layered structure provides a channel for the migration of reaction products, and a quantitative assessment of this effect is the task of the next stages.

5 Methodology

The choice of method is determined by the requirements of the problem. It is necessary to model a three-layer heterostructure of thousands of atoms at 1200 K, where a metallic bond in tungsten and covalent bonds in two-dimensional layers coexist. Direct DFT modeling of such systems is computationally impossible. Classical potentials such as EAM for metals and Tersoff for covalent crystals are good separately but work poorly at interfaces where different bond types meet. Combining them in one system is difficult and often physically incorrect.

Therefore, molecular dynamics with NEP neural network potentials in the GPUMD 5.6 package was chosen. NEP is trained on DFT data and retains near-first-principles accuracy at the speed of classical molecular dynamics.

The main potential is the universal NEP89. It covers 89 elements, including W, B, N, and C, and contains a built-in ZBL potential for the correct description of short-range nuclear repulsion. One potential describes the entire system, eliminating the compatibility problem.

Structures were generated in ASE. Tungsten was built as a BCC lattice with a parameter of 3.18 Å. h-BN and graphene were built as two-dimensional hexagonal lattices with parameters of 2.51 Å and 2.46 Å. The layers were placed along the Z axis and centered in XY relative to the tungsten substrate.

Two dynamic regimes were used for validation: NPT with the MTTK integrator at zero pressure to determine the equilibrium lattice and NVT with the Nose–Hoover thermostat at 1200 K to check thermal stability. Static energies were calculated by steepest-descent minimization.

The integration step was 1 fs. The duration of dynamic calculations ranged from 20 to 20,000 steps depending on the task. The adhesion energy and insertion energy were calculated as differences of the total potential energies of the corresponding systems. Visualization was performed in OVITO. Numerical errors were monitored by the absence of NaN in the output files. Calculations were performed on an NVIDIA GeForce RTX 3060 GPU.

6 Potential Validation

The correctness of any atomistic modeling is determined by the quality of the interatomic potential. No beautiful visualizations or large cells matter if the potential does not reproduce the basic physical properties of materials. Therefore, the first task after assembling the tool is to validate NEP89.

Validation was carried out using six independent parameters: the tungsten lattice parameter, the thermal expansion of tungsten, the C–C bond length in graphene, the B–N bond length in hexagonal boron nitride, the thermal stability of the three-layer system, and the adhesion energy of the W/h-BN interface.

1. Tungsten lattice parameter

The first test checks whether the potential reproduces the equilibrium crystal structure of tungsten. If the equilibrium interatomic distances are described incorrectly, all subsequent energies and configurations will be wrong, and the whole work will lose meaning.

A BCC tungsten cell of 54 atoms was modeled in the isothermal-isobaric NPT ensemble at 300 K and zero external pressure. The cell could change its volume and reach equilibrium.

Result: the lattice parameter was 3.1800 Å. The experimental value is 3.18 Å. The deviation is zero within the calculation accuracy.

The agreement means that NEP89 correctly reproduces the main minimum of the potential energy of tungsten. This is the foundation without which nothing further can be done.

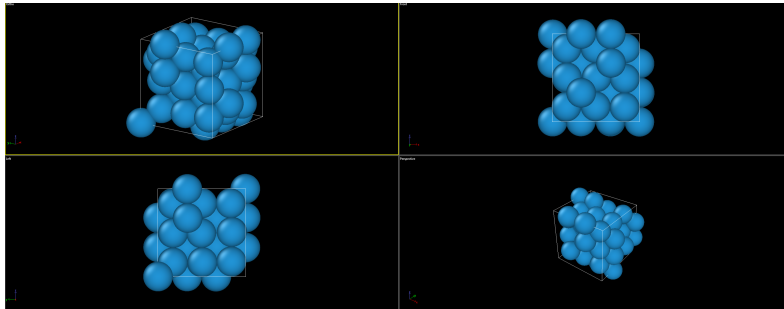


Figure 1: Crystal lattice of tungsten at 300 K. The lattice parameter of 3.1800 Å coincides with the experimental value of 3.18 Å.

2. Thermal expansion of tungsten

The second test checks the behavior of the potential upon heating. Not only the position of the energy minimum is important, but also its asymmetry, that is, the anharmonicity of vibrations. If anharmonicity is not reproduced, the lattice will either remain unchanged upon heating or shrink, which contradicts physics.

The same cell was heated to 1200 K in NPT at zero pressure. After equilibration, the lattice parameter was 3.1843 Å, which is 0.14% greater than the room-temperature value.

This growth is physically correct: upon heating, the vibration amplitude of atoms increases, and the average interatomic distances grow. NEP89 correctly describes not only the equilibrium but also the anharmonicity.

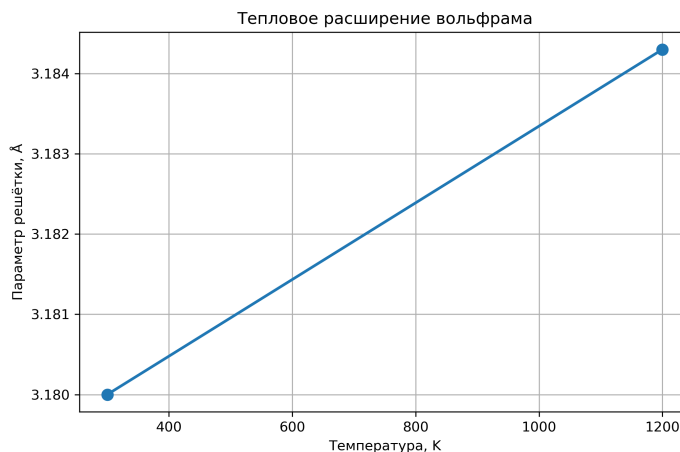


Figure 2: Thermal expansion of tungsten: upon heating from 300 K to 1200 K, the lattice parameter increases from 3.1800 Å to 3.1843 Å.

3. C–C bond length in graphene

The third test concerns the outer protective layer. The barrier properties of graphene are directly related to the integrity and strength of its covalent bonds.

Pure graphene, a hexagonal lattice of 128 carbon atoms, was modeled. After energy minimization, the bond length between neighboring atoms was measured. The value obtained was 1.4203 Å compared with the experimental 1.42 Å. The deviation is 0.02%.

Such high accuracy means that NEP89 is suitable for modeling graphene and its barrier properties.

4. B–N bond length in h-BN

A similar test for hexagonal boron nitride. The stability of h-BN and its behavior upon the introduction of boron reaction products depend on the accuracy of the B–N bond.

Pure h-BN of 128 atoms was modeled. After minimization, the B–N bond was measured. The result was 1.4491 Å compared with the experimental 1.45 Å. The deviation is 0.06%.

The potential correctly describes this layer as well.

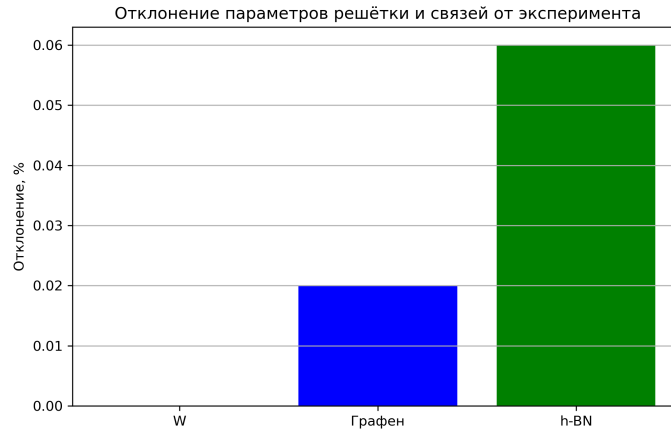


Figure 3: Bond lengths C–C in graphene and B–N in h-BN compared with experimental values.

5. Thermal stability of the three-layer system

After testing individual materials, the system as a whole is checked. The question is simple: does the structure fall apart by itself at the working temperature, before any irradiation.

A three-layer system was assembled: tungsten, 2000 atoms; h-BN, 338 atoms; graphene, 288 atoms. The layers were placed with controlled gaps and centered. The system was heated in NVT at 1200 K for 20,000 steps (20 ps).

The calculation completed without NaN. The layers remained separated, and no mixing was observed. The tungsten lattice remained ordered, and graphene and h-BN stayed in place.

This is the first direct indication that the architecture has the right to exist.

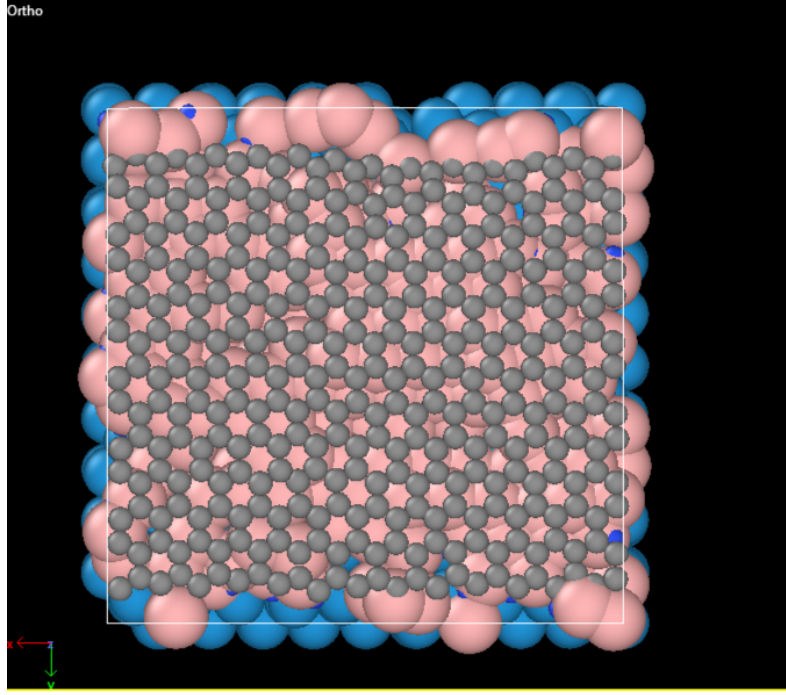


Figure 4: The three-layer W/h-BN/graphene system at 1200 K. The structure remains intact, and the layers do not mix.

6. Adhesion energy of W/h-BN

The last test checks whether tungsten and h-BN attract each other. If the adhesion energy is positive, the layers repel, and the structure is unviable.

The total potential energies of three systems were calculated: the combined W+h-BN system, tungsten alone, and h-BN alone. The adhesion energy is:

$$E_{\text{adh}} = E_{\text{W+BN}} - E_{\text{W}} - E_{\text{BN}} \quad (3)$$

Values:

- $E(\text{W+BN}) = -29353.66 \text{ eV}$;
- $E(\text{W}) = -26426.25 \text{ eV}$;
- $E(\text{BN}) = -2910.93 \text{ eV}$.

Hence:

$$E_{\text{adh}} = -16.48 \text{ eV} \quad (4)$$

The negative sign means that the combined system is more favorable than the separated layers. h-BN is attracted to tungsten and does not tend to detach spontaneously. The value refers to a specific cell and is not recalculated per unit area, but the sign and order of magnitude confirm the stability of the interface.

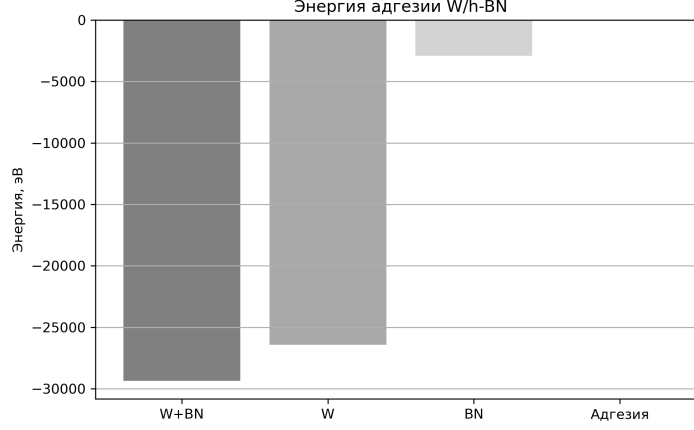


Figure 5: Adhesion energy of the W/h-BN interface. The negative value of -16.48 eV indicates stable interlayer bonding.

The summary validation results are given in Table 1.

Table 1: Validation results for the NEP89 potential

Test	Result	Reference	Deviation
W lattice parameter	3.1800 Å	3.18 Å	0%
W thermal expansion	3.1843 Å	–	+0.14%
C–C bond in graphene	1.4203 Å	1.42 Å	0.02%
B–N bond in h-BN	1.4491 Å	1.45 Å	0.06%
Thermal stability	stable	–	–
W/h-BN adhesion	-16.48 eV	negative	–

Conclusion on validation: NEP89 reproduces the physical properties of tungsten, graphene, and h-BN with high accuracy. The lattice parameters and bond lengths agree with experiment, the thermal expansion is correct, the three-layer system is stable at the working temperature, and the W/h-BN interface is stable. The potential is suitable for modeling the W-B-N-C system.

7 The Helium Potential Problem

After the successful validation of NEP89 for W, B, N, and C, it was logical to check whether the same potential can describe helium. This question is key because helium erosion is considered the main threat to the first wall.

I built a BCC tungsten cell of 54 atoms and placed one He atom at its center. After energy minimization, I calculated the insertion energy:

$$E_{\text{ins}} = E_{\text{W+He}} - E_{\text{W}} = -41.7 \text{ eV} \quad (5)$$

DFT calculations give $+4 \dots +6$ eV. A negative value is qualitatively wrong: NEP89 predicts that helium dissolves favorably in tungsten. In reality, helium is insoluble, and that is exactly why it forms bubbles.

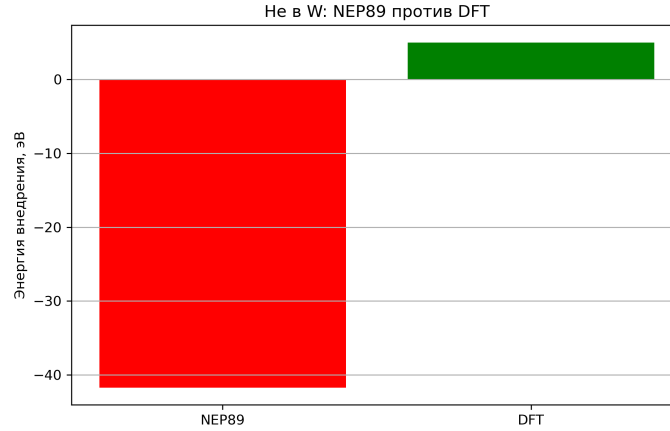


Figure 6: Insertion energy of He in W: NEP89 gives -41.7 eV, while DFT gives $+4 \dots +6$ eV.

This conclusion was confirmed by correspondence with the developer of the NEP method and the GPUMD package, Professor Zheyong Fan. He explained that many-body potentials are not additive: simply combining different NEP models in one simulation is physically incorrect. For the W-B-N-C system, a single potential trained on the whole set of elements is needed. Regarding helium, he said that no reliable ready-made datasets or potentials exist. By his estimate, about 10,000 training configurations are required, of which about 70% can be taken from databases and 30% must be generated additionally. The hardest part is determining which configurations are missing.

This turns the negative test into a documented limitation of the method. The task of creating a He potential is moved to the second part of the work.

8 Limitations

This work is the first stage and has natural limitations.

First, NEP89 does not describe He–W, so helium erosion was not modeled in this work. This is the main limitation because helium is the key destructive factor.

Second, the cells are small: from 54 to several thousand atoms. This is sufficient for validation and thermal-stability checks, but the growth of blisters, fuzz formation, and long-range cascades require systems orders of magnitude larger.

Third, the simulation time is limited to tens of picoseconds. The real wall lifetime is measured in months and years, whereas molecular dynamics covers nanoseconds. The long-term evolution of the coating cannot be assessed.

Fourth, the statistics are limited to single runs. Quantitative estimates with error

bars will require a series of runs with different initial conditions.

Fifth, neutron irradiation was not modeled directly. The $^{10}\text{B}(n, \alpha)^7\text{Li}$ reaction remains a task for the next stages.

Sixth, the thermal contact between layers was considered only through structural stability. How much h-BN worsens heat removal is a separate question.

All these limitations do not cancel the results but outline their boundaries.

9 Future Work

The first priority is to create a potential for helium and to include it in a unified W-B-N-C-He model. After that, modeling of helium erosion and then of the combined action of He and H will become possible.

The second direction is modeling of the $^{10}\text{B}(n, \alpha)^7\text{Li}$ reaction in the h-BN layer: the birth of lithium and helium, their migration, and their influence on the coating integrity.

The third direction is the transition to larger cells and longer times using supercomputer resources.

The fourth is the accumulation of statistics to obtain quantitative estimates.

The fifth is the analysis of thermal contact between layers and, if necessary, optimization of the architecture.

10 Conclusions

In this work, a multilayer protective coating architecture W/h-BN/graphene was proposed, in which each layer performs its own function: tungsten provides heat removal and mechanical strength, h-BN with boron-10 absorbs neutrons, and graphene acts as a barrier for He and H.

Main results:

- a computational approach was selected and justified: molecular dynamics with NEP neural network potentials in GPUMD;
- the universal NEP89 potential was validated for tungsten, graphene, and h-BN: deviations of 0%, 0.02%, and 0.06%;
- the three-layer system is stable at 1200 K;
- the W/h-BN adhesion energy is negative: -16.48 eV;
- it was established that NEP89 is unsuitable for He-W, and the creation of a He potential was identified as a separate task.

The work confirms the physical validity of the proposed architecture and the suitability of the chosen tool for the W-B-N-C system. It lays the foundation for further modeling of helium erosion, neutron irradiation, and a full-scale calculation of the coating.

Acknowledgments

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